

Report of PhD activities in the academic year
2017 / 2018

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Abstract

This document describes the achievements, current state and future plans of the research project, moreover it contains the courses and conferences attended in the 2017 / 2018 academic year.

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1 Current state of the research topic - Atmosphere modelling

1.1 The shallow domain concept

- Natural, large-scale geophysical fluids share the shallow domain characteristics
- "almost" two-dimensional set $\rightarrow$ we can significantly simplify the model in the vertical direction
- a domain that is excessively greater horizontally than in the vertical dimension $\rightarrow$ leading to the classic hydrostatic approximation of the Navier-Stokes equations

The main ideas and tools in this scenario:

- introduce the aspect ratio $\epsilon = \frac{\text{characteristic depth}}{\text{characteristic width}}$,
- anisotropic viscosity hypothesis,
- rescaling, $x = x_1, y = x_2, z = \epsilon x_3$.

Correspondingly, we rescale

- the kinematic terms: $v_3 = \epsilon u_3^\epsilon$,
- the viscosity terms: $\nu_z = \epsilon^2 \nu_3$,
- the diffusivity terms: $K_x = \epsilon K_1, K_y = K_2, K_z = \epsilon^2 K_3$.

1.2 Modelling - challenges and choices

- Coordinate system $\rightarrow$ we fix it in a way so that x and y match the downwind and crosswind direction, respectively (while z represents the vertical dimension).

Natural assumption in the downwind direction: diffusion is negligible compared to downwind transport, i.e.

$$|u_1 \partial_{x_1} C| \gg \left| K_x \frac{\partial^2 C}{\partial x_1^2} \right|,$$

so we can take advantage of the additional scaling equation $K_x = \epsilon K_1$.

- Domain $\rightarrow$ we use locally acceptable Cartesian coordinates — this allows us to ignore large-scale effects such as the curvature of the Earth.

Challenges: local area model $\rightarrow$ open boundary conditions on the lateral boundary. We use a fix zero approach — the concentration being set to zero actually stands for a constant concentration with the background flow being removed.

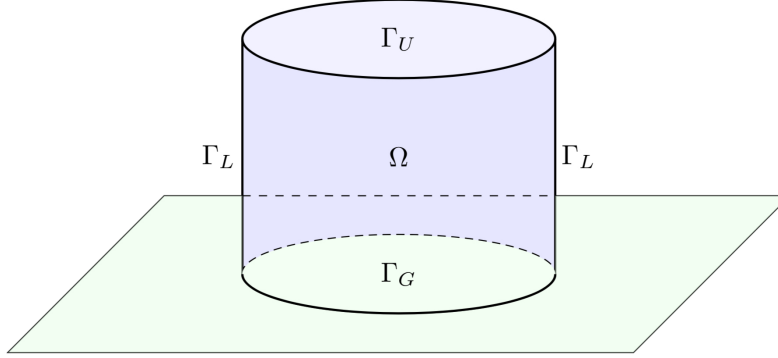


Figure 1: A local slice of the atmosphere

- Source function of the concentration $\rightarrow$ Gaussian impulse approach. We use a parametrized nascent delta function to describe the source term, with the parameter being ϵ . We have $S_\epsilon(t, t_s, x, x_s) = s(t, t_s)\delta_\epsilon(x - x_s)$, with the activation term $s(t, t_s) = IH(t - t_s)$ where $H(\cdot)$ is the Heaviside step function. In the limit model we naturally arrive to the delta distribution as a source.

1.3 The anisotropic Navier-Stokes equations

We use the incompressible Navier-Stokes equations in all three space dimensions for the fluid velocity.

$$\partial_t u_1^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_1^\epsilon - \Delta_\nu u_1^\epsilon - \alpha u_2^\epsilon + \epsilon \beta u_3^\epsilon + \partial_1 p^\epsilon = 0 \text{ in } \Omega \times (0, T)$$

$$\partial_t u_2^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_2^\epsilon - \Delta_\nu u_2^\epsilon + \alpha u_1^\epsilon + \partial_2 p^\epsilon = 0 \text{ in } \Omega \times (0, T)$$

$$\epsilon^2 \{ \partial_t u_3^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_3^\epsilon - \Delta_\nu u_3^\epsilon \} - \epsilon \beta u_1^\epsilon + \partial_3 p^\epsilon = 0 \text{ in } \Omega \times (0, T)$$

$$\nabla \cdot \mathbf{u}^\epsilon = 0 \text{ in } \Omega \times (0, T)$$

$$\partial_t C^\epsilon + \mathbf{u}^\epsilon \cdot \nabla C^\epsilon = \epsilon K_1 \frac{\partial^2 C^\epsilon}{\partial x_1^2} + K_2 \frac{\partial^2 C^\epsilon}{\partial x_2^2} + K_3 \frac{\partial^2 C^\epsilon}{\partial x_3^2} + S_\epsilon \text{ in } \Omega \times (0, T)$$

$$\mathbf{u}^\epsilon = 0 \text{ on } \Gamma \times (0, T)$$

$$\mathbf{u}^\epsilon(\cdot, t = 0) = \mathbf{u}_0, C^\epsilon(\cdot, t = 0) = C_0^\epsilon \text{ in } \Omega$$

$$C^\epsilon = 0 \text{ on } (\Gamma_U \cup \Gamma_L) \times (0, T), \partial_2 C^\epsilon = \partial_3 C^\epsilon = 0 \text{ on } \Gamma_G \times (0, T), \partial_1 C^\epsilon \in L^2 L^2(0, T; \Gamma_G)$$

1.4 The limit model

In the vertical dimension we arrive to the hydrostatic approximation.

$$\partial_t u_1 + \mathbf{u} \cdot \nabla u_1 - \Delta_\nu u_1 - \alpha u_2 + \partial_1 p = 0 \text{ in } \Omega \times (0, T)$$

$$\partial_t u_2 + \mathbf{u} \cdot \nabla u_2 - \Delta_\nu u_2 - \alpha u_1 + \partial_2 p = 0 \text{ in } \Omega \times (0, T)$$

$$\partial_3 p = 0 \text{ in } \Omega \times (0, T)$$

$$\nabla \cdot \mathbf{u} = 0 \text{ in } \Omega \times (0, T)$$

$$\partial_t C + \mathbf{u} \cdot \nabla C = K_2 \frac{\partial^2 C}{\partial x_2^2} + K_3 \frac{\partial^2 C}{\partial x_3^2} + S \text{ in } \Omega \times (0, T)$$

$$u_1 = u_2 = u_3 n_3 = 0 \text{ on } \Gamma \times (0, T)$$

$$u_i(\cdot, t = 0) = u_{0i}, C(\cdot, t = 0) = C_0 \text{ in } \Omega, i = 1, 2$$

$$C = 0 \text{ on } (\Gamma_U \cup \Gamma_L) \times (0, T), \partial_2 C = \partial_3 C = 0 \text{ on } \Gamma_G \times (0, T), \partial_1 C \in L^2 L^2(0, T; \Gamma_G)$$

1.5 Main result

Let $\mathbf{u}_0 \in L^2(\Omega)^3$, with $\nabla \cdot \mathbf{u}_0 = 0$, $\mathbf{u}_0 \cdot \mathbf{n} = 0$ on $\partial\Omega$; and let $C(0) \in L^2(\Omega)^3$, $C(0) = 0$ on $\partial\Omega$; there exists a weak solution $(\mathbf{u}, C)$ of the hydrostatic equations of the polluted atmosphere, obtained as a limit of weak solutions $(\mathbf{u}^\epsilon, C^\epsilon)$ of the anisotropic equations as the aspect ratio ϵ tends to 0.

1.6 Proof

A priori estimates, weak convergence

The main steps for proving the weak convergence are the following.

- Energy inequality.

$$\begin{aligned}
& \frac{1}{2} (\|\mathbf{u}_H^\epsilon(t)\|_{L^2}^2 + \epsilon^2 \|u_3^\epsilon(t)\|_{L^2}^2 + \|C^\epsilon(t)\|_{L^2}^2) + \int_0^t [\|\nabla_\nu \mathbf{u}_H^\epsilon\|_{L^2}^2 + \epsilon^2 \|\nabla_\nu u_3^\epsilon\|_{L^2}^2] d\tau \\
& + \int_0^t \left[(\epsilon K_1 - \zeta \frac{\epsilon}{2} K_1 c_T (c_S + 1)) \|\partial_1 C^\epsilon\|_{L^2}^2 + (K_2 - \zeta \frac{\epsilon}{2} K_1 c_T (c_S + 1)) \|\partial_2 C^\epsilon\|_{L^2}^2 \right. \\
& \left. + (K_3 - \zeta \frac{\epsilon}{2} K_1 c_T (c_S + 1)) \|\partial_3 C^\epsilon\|_{L^2}^2 \right] d\tau \\
& \leq \frac{1}{\zeta} K_1 \frac{\epsilon}{2} \int_0^t \|\partial_1 C^\epsilon\|_{L^2(\Gamma_G)}^2 + \frac{1}{2} (\|\mathbf{u}_H^\epsilon(0)\|_{L^2}^2 + \epsilon^2 \|u_3^\epsilon(0)\|_{L^2}^2 + \|C^\epsilon(0)\|_{L^2}^2) \\
& + \int_0^t (S^\epsilon, C^\epsilon) d\tau.
\end{aligned} \tag{1}$$

- A priori estimates: $L_t^\infty L_x^2$ boundedness for C^ϵ ; $L_t^2 L_x^2$ boundedness for $\sqrt{\epsilon} \partial_1 C^\epsilon, \partial_2 C^\epsilon, \partial_3 C^\epsilon$.
- Weak convergence of the linear terms.
- Weak convergence of nonlinear terms.
 - Nonlinear terms not including the C^ϵ term - divergence free condition
 - The term $(\mathbf{u}^\epsilon C^\epsilon, \nabla \tilde{C})$.

We need a compactness result to obtain $u_i^\epsilon C^\epsilon \rightharpoonup u_i C$.

For this we show that

- we have a bound for the time derivative of the concentration:

$$\frac{\partial C^\epsilon}{\partial t} = -u^\epsilon \nabla C^\epsilon + \sqrt{\epsilon} K_1 \partial_1 \partial_1 \sqrt{\epsilon} C^\epsilon + K_2 \partial_2 \partial_2 C^\epsilon + K_3 \partial_3 \partial_3 C^\epsilon - S^\epsilon,$$

Eventually we have $\frac{\partial C}{\partial t} \in L^1 W^{-1,1}$,

- we have an equicontinuity for velocity: $\|u_i^n - u_i^n(\cdot + \xi, t)\|_{L^2 L^2} \rightarrow 0$ as $\xi \rightarrow 0$.

Combining these results allow us to apply a compactness theorem that provides the convergence.

- We also provide an existence theorem for the extended system.
- We have generalized the main theorem for the seaside case, when there is interaction between water and air $\rightarrow$ different boundary conditions.

1.7 Outlook

Convergence, diffusion matrix, FEM.

2 Courses, summer schools and seminars

2.1 Courses

2.1.1 Physics of the atmosphere - Prof G. Curci, Prof. G. Redaelli, University of L'Aquila

This course gave an overview of selected topics related to the dynamics of the atmosphere. The general characteristics of the atmosphere were described in terms of mass and chemical composition. Gases with the highest mixing ratio are nitrogen, oxygen and argon - it is pointed out that these are constants, and the variabilities are provided by the remainder gases which are present in much smaller percentage: trace gases and aerosols. Although their lifetime is small, these later's effect is huge as they are able to capture radiation and alter energy balance. A big chapter in this course is represented by the thermodynamics of the atmosphere and gas laws. The ideal gas equation is considered in different forms, as well as the concept of hydrostatic balance and geopotential. The next step is the first law of thermodynamics in different approaches, and the notion of an adiabatic process. The main ideas and building blocks of chemical and photochemical reactions (thermal decomposition, photolysis) in the atmosphere are introduced as well as the notion of the rate of reaction, and several chemical reaction equations are solved in this chapter.

2.1.2 Numerical methods for hyperbolic problems - Prof. G. Russo (University of Catania), GSSI

This course gave an overview on shock capturing schemes for the numerical solution of systems of conservation and balance laws. The emphasis was on the construction of high order finite volume and finite difference schemes, which are based on the three main ingredients: numerical flux, high order non oscillatory reconstruction, and high order integration in time. The first part of the course illustrated some models in which the nonlinearity produces shocks that propagate in time. Simple three point schemes, such as upwind, Lax-Friedrichs, and Lax-Wendroff was reviewed, and their behaviour on linear equations and systems analysed, before the illustration of more complex methods for the quasi-linear systems. The step by step construction of the schemes was illustrated in lab sessions, and the methods were applied to concrete examples of physical interest. The last part of the course was devoted to systems with source term.

2.2 Conferences, seminars, summer schools

2.2.1 Intensive Program on Fluids and Waves, GSSI

This training activity within the European Innovating Training Network Mod-CompShock surveyed recent results in the analysis of fluids and waves with emphasis on modelling and computational methods as well as its theoretical aspects.

- a) Week 1. Workshop/School in Numerics for Fluids and Waves
- b) Week 2. Mini courses by J. Bedrossian. and D. Cordoba
- c) Week 3. Mini courses by C. Bardos, Z. Grujic, J. Malek
- d) Week 4. Workshop on Mathematical fine structures in fluid dynamics

2.2.2 Women in applied and computational Mathematics, GSSI

This three-day event was designed for researchers challenged by the application of various techniques in modern analysis and computational mathematics to intriguing problems arising in physics, material science and biology. The topic and the results of this present research project were represented - together with the work of many young female researchers - in the poster session.

2.2.3 Evans function methods in the stability analysis of periodic wavetrains, Prof. R. Plaza

This lecture series provided an introduction to recent techniques in the spectral stability analysis of spatially periodic travelling waves which involve the periodic Evans function introduced by Gardner.

2.2.4 On the spectral stability of traveling fronts for reaction diffusion-degenerate equations, Prof. R. Plaza

2.2.5 Secondment at the University of Sussex, Prof. Omar Lakkis

The aim of this 6-week secondment was to create both the theoretical and technical foundations for numerical simulations in order to visualise the results of the project. The main results and steps are the following:

- a) Software packages such as Docker, Anaconda, ParaView, FEniCS and Alberta have been installed
- b) Writing scripts to login / connect to the HPC server at the Sussex University
- c) Getting to know FEniCS: tutorial files and the FEniCS version of the "hello world" programs
- d) Creating a basis for future work with specific theoretical details that arise when implementing this project - such as the loss of incompressibility when it comes to the FEM approach, and the use of Lagrange multipliers.

2.2.6 Waves in Flows Summer School in Prague

The aim of this summer school was to present a comprehensive series of lectures on various aspects of waves and their role in mathematical analysis and numerical simulation of different types of flows. The school was organised as a multidisciplinary event with topics ranging from theoretical and numerical mathematics up to physics and applications. The emphasise was on models related to topics originating in environmental, biomedical and industrial applications. Besides of the course lectures, the program included a one day workshop dedicated to short lectures given by attending scientists.

References

- [Adams and Fournier, 2003] Adams, R. A. and Fournier, J. J. (2003). *Sobolev spaces*, volume 140. Academic press.
- [Ali et al., 1998] Ali, I., Kalla, S., and Khajah, H. (1998). A partial differential equation related to a problem in atmospheric pollution. *Mathematical and computer modelling*, 28(12):1–6.
- [Anderson and Wendt, 1995] Anderson, J. D. and Wendt, J. (1995). *Computational fluid dynamics*, volume 206. Springer.
- [Andria et al., 2000] Andria, G., Lay-Ekuakille, A., and Notarnicola, M. (2000). Mathematical models for atmospheric and industrial pollutant prediction. In *XVI IMEKO World Congress, Vienna, Austria*.
- [Arya, 2001] Arya, P. S. (2001). *Introduction to micrometeorology*, volume 79. Elsevier.
- [Azérad and Guillén, 2001] Azérad, P. and Guillén, F. (2001). Mathematical justification of the hydrostatic approximation in the primitive equations of geophysical fluid dynamics. *SIAM Journal on Mathematical Analysis*, 33(4):847–859.
- [Besson and Laydi, 1992] Besson, O. and Laydi, M. (1992). Some estimates for the anisotropic navier-stokes equations and for the hydrostatic approximation. *ESAIM: Mathematical Modelling and Numerical Analysis*, 26(7):855–865.
- [de Swart, 2016] de Swart, H. E. (2016). The governing equations and the dominant balances of flow in the atmosphere and ocean. *USPC Summerschool, Utrecht*.
- [Goyal and Kumar, 2011] Goyal, P. and Kumar, A. (2011). Mathematical modeling of air pollutants: An application to indian urban city. In *Air Quality-Models and Applications*. InTech.
- [Hosseini, 2013] Hosseini, B. (2013). Dispersion of pollutants in the atmosphere: a numerical study. Master’s thesis, Simom Fraser University.

- [Kathirgamanathan et al., 2002] Kathirgamanathan, P., McKibbin, R., and McLachlan, R. (2002). Source term estimation of pollution from an instantaneous point source. *Research Letters in Information Mathematic Science*, Vol. 3, pp. 59-67.
- [Lions, 1996] Lions, P.-L. (1996). *Mathematical Topics in Fluid Mechanics: Volume 2: Compressible Models*, volume 2. Oxford University Press on Demand.
- [Marsaleix et al., 2006] Marsaleix, P., Auclair, F., and Estournel, C. (2006). Considerations on open boundary conditions for regional and coastal ocean models. *Journal of Atmospheric and Oceanic Technology*, 23(11):1604–1613.
- [Monin, 1959] Monin, A. (1959). On the boundary condition on the earth surface for diffusing pollution. In *Advances in Geophysics*, volume 6, pages 435–436. Elsevier.
- [Pedlosky, 2013] Pedlosky, J. (2013). *Geophysical fluid dynamics*. Springer Science & Business Media.
- [Prodanova et al., 2008] Prodanova, M., Perez, J. L., Syrakov, D., San Jose, R., Ganev, K., Miloshev, N., and Roglev, S. (2008). Application of mathematical models to simulate an extreme air pollution episode in the bulgarian city of stara zagora. *Applied Mathematical Modelling*, 32(8):1607–1619.
- [Simon, 1986] Simon, J. (1986). Compact sets in the space $l^p(0, t; b)$. *Annali di Matematica pura ed applicata*, 146(1):65–96.
- [Temam, 2001] Temam, R. (2001). *Navier-Stokes equations: theory and numerical analysis*, volume 343. American Mathematical Soc.
- [Temam and Ziane, 2005] Temam, R. and Ziane, M. (2005). Some mathematical problems in geophysical fluid dynamics. *Handbook of mathematical fluid dynamics*, 3:535–658.
- [Wang et al., 2014] Wang, Q., Zhou, W., Wang, D., and Dong, D. (2014). Ocean model open boundary conditions with volume, heat and salinity conservation constraints. *Advances in Atmospheric Sciences*, 31(1):188–196.
- [Yeh and Huang, 1975] Yeh, G.-T. and Huang, C.-H. (1975). Three-dimensional air pollutant modeling in the lower atmosphere. *Boundary-Layer Meteorology*, 9(4):381–390.
- [Zhuk et al., 2016] Zhuk, S., Tchrakian, T. T., Moore, S., Ordóñez-Hurtado, R., and Shorten, R. (2016). On source-term parameter estimation for linear advection-diffusion equations with uncertain coefficients. *SIAM Journal on Scientific Computing*, 38(4):A2334–A2356.